

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 5

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Given:

- Total tosses (n) = 10
- Outcomes: H H T H H H T H H T
- Number of Heads (x) = 7

Formula:

$$\hat{p} = \frac{x}{n} \quad \hat{p} = \frac{7}{10} = 0.7$$

Calculation:

$$\hat{p} = \frac{7}{10} = 0.7$$

Answer:

The MLE estimate of the probability of getting **Heads** is:

$$\hat{p} = 0.7 \quad \boxed{\hat{p} = 0.7}$$

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Given:

- Total students = 50
- Students passed = 42

Formula:

$$\hat{p} = \frac{x}{n} \quad \hat{p} = \frac{42}{50} = 0.84$$

Calculation:

$$\hat{p} = \frac{42}{50} = 0.84$$

Answer:

$$\hat{p} = 0.84 \quad \boxed{\hat{p} = 0.84}$$

The MLE estimate of the probability that a student passes is **0.84 (84%)**.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.

Given:

- Total bulbs = 100
- Defective bulbs = 8

Formula:

$$\hat{p} = \frac{x}{n} \quad \hat{p} = \frac{8}{100} = 0.08$$

Calculation:

$$\hat{p} = \frac{8}{100} = 0.08 \quad \hat{p} = \frac{8}{100} = 0.08$$

Answer:

$$\hat{p} = 0.08 \quad \boxed{\hat{p} = 0.08} \quad \hat{p} = 0.08$$

The MLE estimate of the probability that a bulb is defective is **0.08 (8%)**.

4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.

Given:

- Total rolls = 60
- Number of times 6 appears = 18

Formula:

$$\hat{p} = \frac{x}{n} \quad \hat{p} = \frac{18}{60} = 0.3$$

Calculation:

$$\hat{p} = \frac{18}{60} = 0.3 \quad \hat{p} = \frac{18}{60} = 0.3$$

Answer:

$$\hat{p} = 0.3 \quad \boxed{\hat{p} = 0.3} \quad \hat{p} = 0.3$$

The MLE estimate of the probability of rolling a 6 is **0.30 (30%)**.

5. Given:

$$\text{Input } (x) = 4$$

$$\text{Weight } (w) = 2$$

$$\text{Actual Output} = 10$$

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Given:

- Input (x) = 4
 - Weight (w) = 2
 - Actual Output = 10
 - Learning Rate (α) = 0.01
 - Gradient = 4
-

(i) Predicted Output

Formula:

$$y = w \times x \quad y = 2 \times 4 = 8$$

Calculation:

$$y = 2 \times 4 = 8 \quad y = 2 \times 4 = 8$$

Predicted Output:

$$8$$

(ii) Calculate the Error

Formula:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output} \quad \text{Error} = 10 - 8 = 2$$

Calculation:

$$\text{Error} = 10 - 8 = 2 \quad \text{Error} = 10 - 8 = 2$$

Error:

$$2$$

(iii) Updated Weight

Formula:

$$w_{\text{new}} = w - \alpha \times \text{Gradient} \quad w_{\text{new}} = 2 - 0.01 \times 4 = 1.96$$

Calculation:

$$w_{\text{new}} = 2 - (0.01 \times 4) = 2 - 0.04 = 1.96 \quad w_{\text{new}} = 2 - 0.04 = 1.96$$

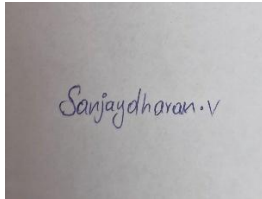
Updated Weight:

$$1.96$$

Code of Conduct:

I Sanjay dharan v (192425002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature
of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation